

Basics of Agriculture

HuNu2073



**BETTER
PRODUCTION**



**BETTER
NUTRITION**



**BETTER
ENVIRONMENT**



**BETTER
LIFE**



1. NATURAL RESOURCES AND ECOSYSTEM

“Caring for Earth is caring for our self and our community” Daniel Christian Wahl

“Healing the land is healing ourselves.” Adrianna Acosta

“Nature and biodiversity are rising up the business agenda faster than anything else”. Steve Kenzie from the UN Global Compact Network

1.1. Introduction to Natural Resources

- A resource is a form of energy and/or matter which is essential for the functioning of living organisms (population and ecosystem). In other words, any part of the environment such as land, water, air, minerals, forest, wild life etc. that humans can utilize to promote their welfare may be regarded as resources.
- A natural capital /resource/ is any asset that can be obtained from the environment e.g. water, soil, plants, wind, animals, minerals, sunlight or energy, etc.
- Soil and water underpin the whole production system. Biodiversity which includes a wide diversity of genetic resources also underpins vibrant food systems.

1.1.1. Overexploitation

- Even being renewable does not mean that the resources last forever. If the renewable resources in a particular area are overexploited for a long period of time, it is entirely possible that they may become endangered or even disappear altogether.
- Overexploitation of natural resources compromises the availability and fair distribution of resources and their associated benefits to the health of ecosystems and the livelihoods and wellbeing of people.

1.1.2. Ecosystem

What is an ecosystem?

What is an Ecosystem Structure?

The **structure** of an **ecosystem** is essentially a description of an environment's organisms and physical features.

It is described under two major headings: Abiotic components and Biotic components.

1.1.2.1. Ecosystem services

- Ecosystem services are the direct and indirect contributions of ecosystems, mostly NRs, to human well-being. Healthy, bio-diverse ecosystems sustain humanity by providing air, water and other essential services.
- Many (ecosystem) goods and services - e.g. wildlife habitat and diversity, watershed services, carbon storage, and scenic landscapes - are traditionally viewed as free benefits to society, or 'public goods'. Lacking a formal market, natural assets are traditionally absent from society's balance sheet; their critical contributions are often overlooked in public, corporate, and individual decision-making.
- The degradation of ecosystems has major consequences for our own health and wellbeing. Healthy ecosystems provide us with a wealth of ecosystem goods and services such as water purification, crop pollination, flood protection, **carbon storage and sequestration**. Their monetary value is considerable and virtually irreplaceable.

... 1.1.2.1. Ecosystem services



1.1.2.2. Ecosystem health and restoration

- Ecosystem restoration means assisting in the recovery of ecosystems that have been degraded or destroyed, as well as conserving the ecosystems that are still intact. Healthier ecosystems, with richer biodiversity, yield greater benefits such as more fertile soils, bigger yields of timber and fish, and larger stores of greenhouse gases.
- Restoration can happen in many ways – for example through actively planting or by removing pressures so that nature can recover on its own. It is not always possible – or desirable – to return an ecosystem to its original state.

1.2. Soils

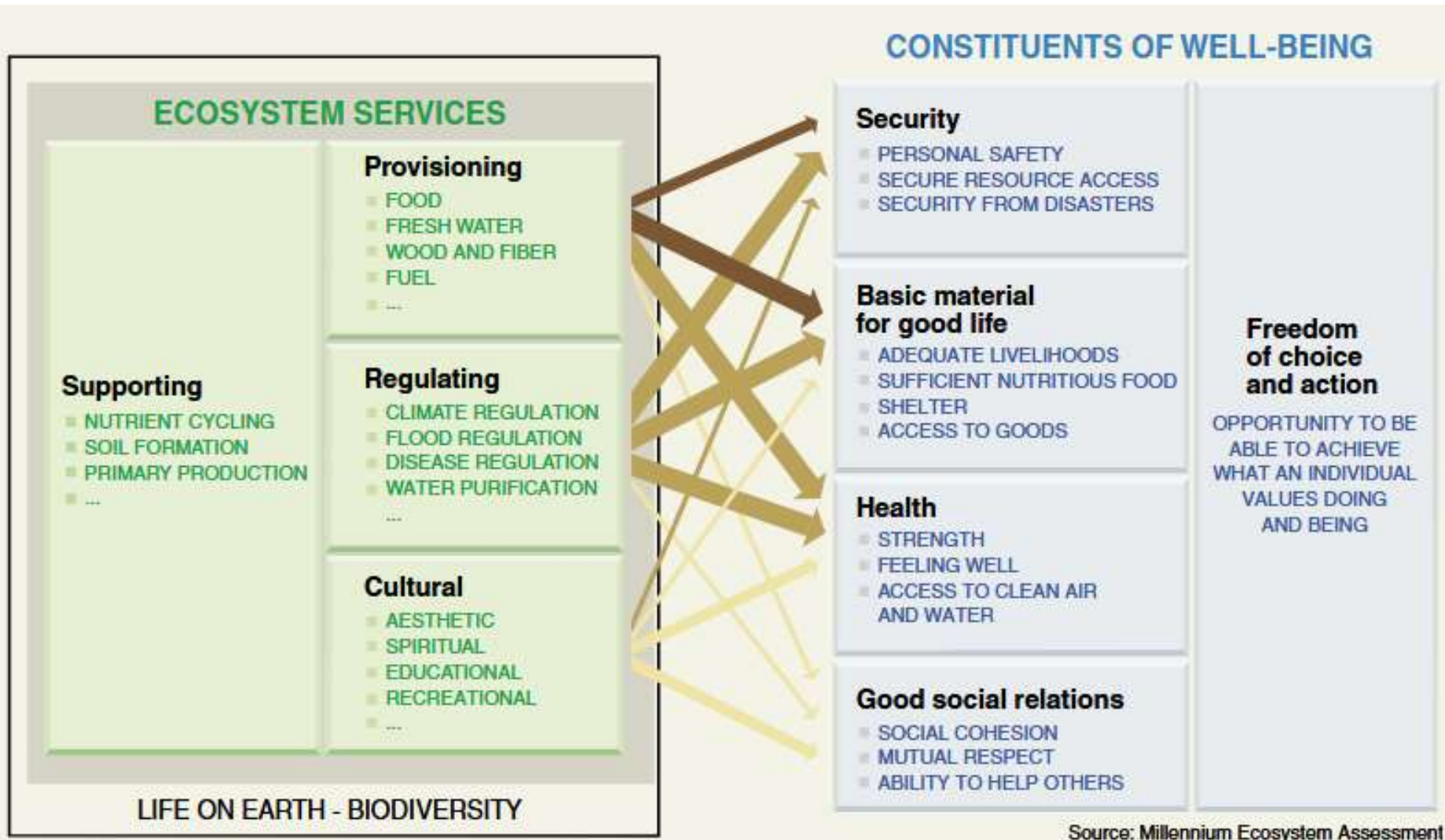


- Soils are fundamental to life on earth, and the ability of soils to provide safe and nutritious food is a key contribution to us and to nature as a whole.
- Soils are the foundation of our food security.
- Food starts with soils. About 95 percent of our food nutrients come from soils, which have a natural capacity to provide nutrients to support crop growth.

...1.2. Soils

- Recently soil has been simply defined as:
The layer(s) of generally loose mineral and/or organic material at or near the planetary surface and supports plants.
- Soil forms so slowly that we should regard it as a non-renewable resource.
- Healthy soil , full of biological life, implies that all aspects of the soil - chemical, physical and biological - are in a perfect status to be able to produce a healthy food. All people depend on nature and ecosystem services to provide the conditions for a decent, healthy, and secure life (Figure 1). Soil functions are at the core of virtually all processes. **Soil Ecosystem services:** Soil provides, regulates and supports a number of ecosystem services including cultural values.
- Soils are in danger of being the 'orphan' asset, forgotten when food security or forests are discussed. The global decline in soil quality is therefore a major concern.

Figure 1. Linkages between Ecosystem Services and Human Well-being



...1.2.1. Major Components of Soil

Mineral soils consist of four major components. These are:

- Mineral (inorganic) constituent: extremely variable in size ranging from 0.0002 to 2 mm in diameter. Minerals, being the major component of soil, exhibit markedly different and diverse effects on the physico-chemical behaviors of soils.
- Organic matter: represents an accumulation of partially decayed and partially synthesized plant and animal residues.
- Soil water: makes up soil solution, together with its dissolved salts, which is important as a medium for supplying nutrients to growing plants.
- Soil air: The amount and composition of soil air is determined to a large degree by the water content of the soil.

... 1.2.1. **Major** Components of Soil

Figure 2. Volume composition of a favorable /an ideal/ soil for plant growth



... 1.2.2. Soil formation

Soils are formed from rocks by weathering. The major parent material for soils are rocks. Soil formation takes place after a gradual exposure of rocks to the elements and activity of organism – called weathering.

Weathering includes

- physical weathering/disintegration - breaking or grinding particles to smaller sizes without changing their chemical composition.
- chemical weathering/decomposition - the dissolving or chemical alteration of minerals

1.2.2. Soil formation



Many types of plants have adapted to grow on rocks or new soil.

1.2.2.1. Agents

in

Physical Weathering

- Temperature fluctuations
- water/ice
- wind,
- gravity
- mechanical actions of animals, and
- mechanical forces of growing plants

Chemical Weathering

- **Oxygen**: in oxidation-reduction reactions
- **water**: in hydrolysis, hydration and (dis)solution reactions
- **carbon dioxide**: in carbonation
- **organic and inorganic acids**
- **bases**, and
- **salts**

1.2.2.2. Processes in Soil Chemical Weathering

Hydrolysis: is the reaction between the H^+ and/or OH^- ions of water and the elements or ions of a mineral.

Oxidation-reduction reactions: Another chemical process involves the simultaneous loss and gain of electrons in substances.

Hydration: is the binding of water molecules to a mineral.

Carbonation The majority of soil organisms release carbon dioxide (CO_2) from respiration. Carbonation is a reaction by which water dissolves carbon dioxide and the dissolved carbon dioxide forms carbonic acid.

1.2.2.3. Factors of Soil Formation/Soil forming factors

The five soil-forming factors that influence the development of soil are:

- Parent Material
- Topography or position in the landscape
- Climate
- Living organisms, especially vegetation
- Time

1.3. Soils and Plants

In agriculture, using manure/fertilizer returns some organic matter to the soil. Any plant, even annual or perennial weeds, drop leaves and twigs into the soil, which becomes food for soil biota. At the same time, their shade reduces solar radiation from killing microorganisms. Therefore, they play a key role in improving soil fertility and making soil conducive for soil biota to thrive.

1.3.1. Essential Plant Nutrients

The criteria to consider an element to be essential are:

- A deficiency of an essential nutrient makes it impossible for the plant to complete the vegetative or reproductive stage of its life cycle.
- Such deficiency is specific to the element in question and can be prevented or corrected only by supplying this element.
- The element is involved directly in the metabolism of the plant.

...1.3.1. Essential Plant Nutrients

- Sixteen elements are known to be essential in plant growth. They can be grouped based on the amount they are required by plants and based on their sources. Eighteen nutrients are necessary for human health.

Macronutrients: nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulfur (S).

Micronutrients: boron (B), zinc (Zn), manganese (Mn), iron (Fe), copper (Cu), molybdenum (Mo), chlorine (Cl).

- Though the majority of the elements come from the soil, they make up very small part of the plant tissue as plant tissues are mainly made of C, H and O (94-99.5%). C, H and O are not limiting under normal condition as they are supplied by the atmosphere and H_2O .

...1.3.1. Essential Plant Nutrients

A healthy soil is a chemically balanced soil. The macronutrient essential soil nutrients, very important for plant growth, are.

1.3.1.1. Primary Nutrients

🌿 Nitrogen is the major nutrient for strong, vigorous growth, dark green leaf color, and photosynthesis.

🌱 Phosphorus helps roots for strong root growth and development.

🌸 Potassium helps flowers and fruits to develop, improving the overall health and strength of the plant.

1.3.1.2. Secondary Nutrients

Calcium is important for general plant vigor.

Magnesium helps regulate the uptake of other plant foods.

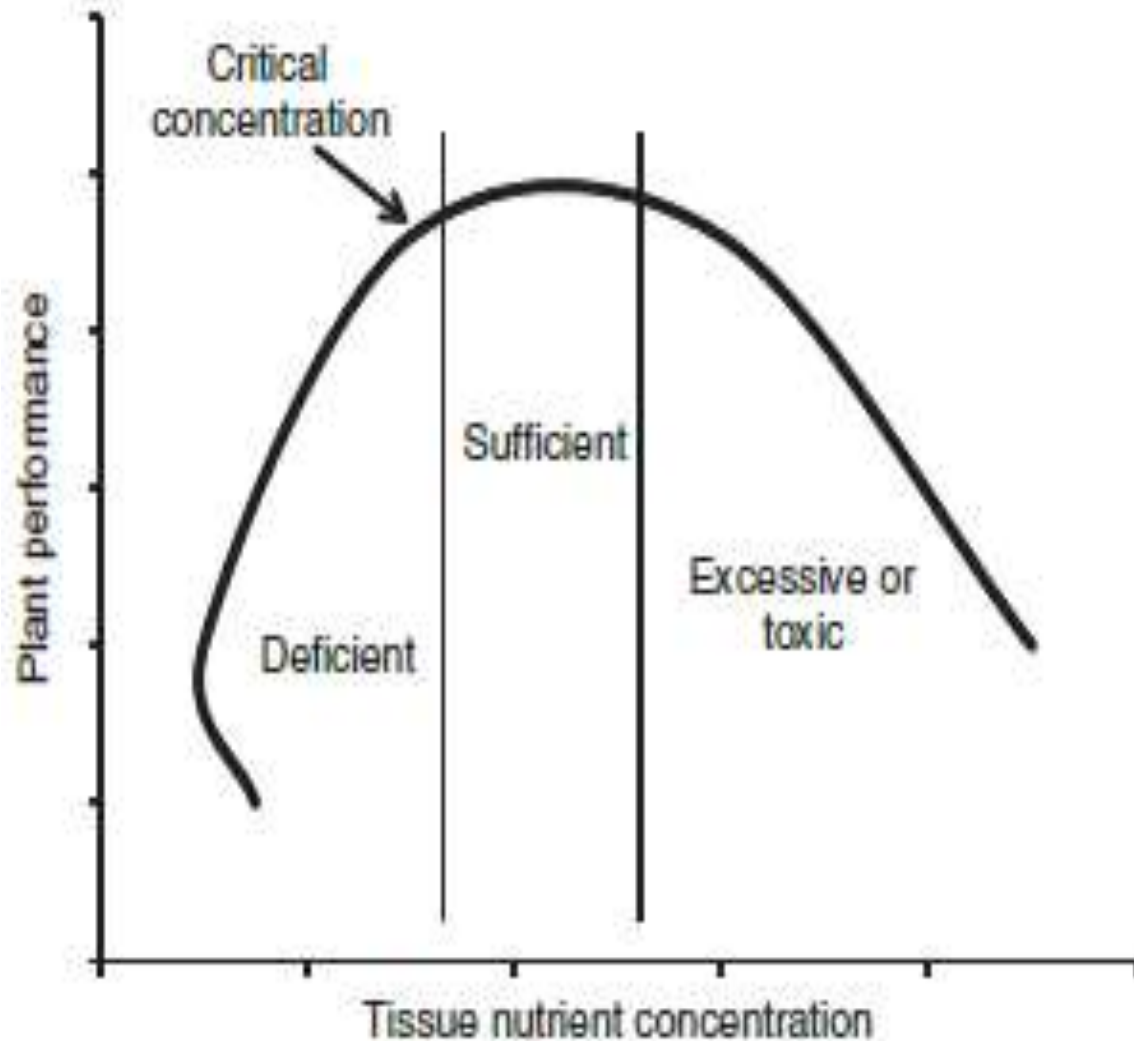
Sulfur helps maintain a dark green color while encouraging more vigorous plant growth.

...1.3.1. Essential Plant Nutrients

- Five additional elements – sodium (Na), cobalt (Co), vanadium (Va), nickel (Ni) and silicon (Si) are essential elements for some plants. Plants absorb many non-essential elements. The content of the micronutrients is also often far in excess of physiological requirements.
- Nutrients are desirable in particular concentrations and can be detrimental to plant growth when concentrations in soil solution are either too low (limiting) or too high (toxicity) [Figure 3].
- The lack of essential nutrients, including macro and micronutrients, leads to the underdevelopment of plants and a decrease in yield and in crop nutritional value.

Nutrient Uptake by Plants

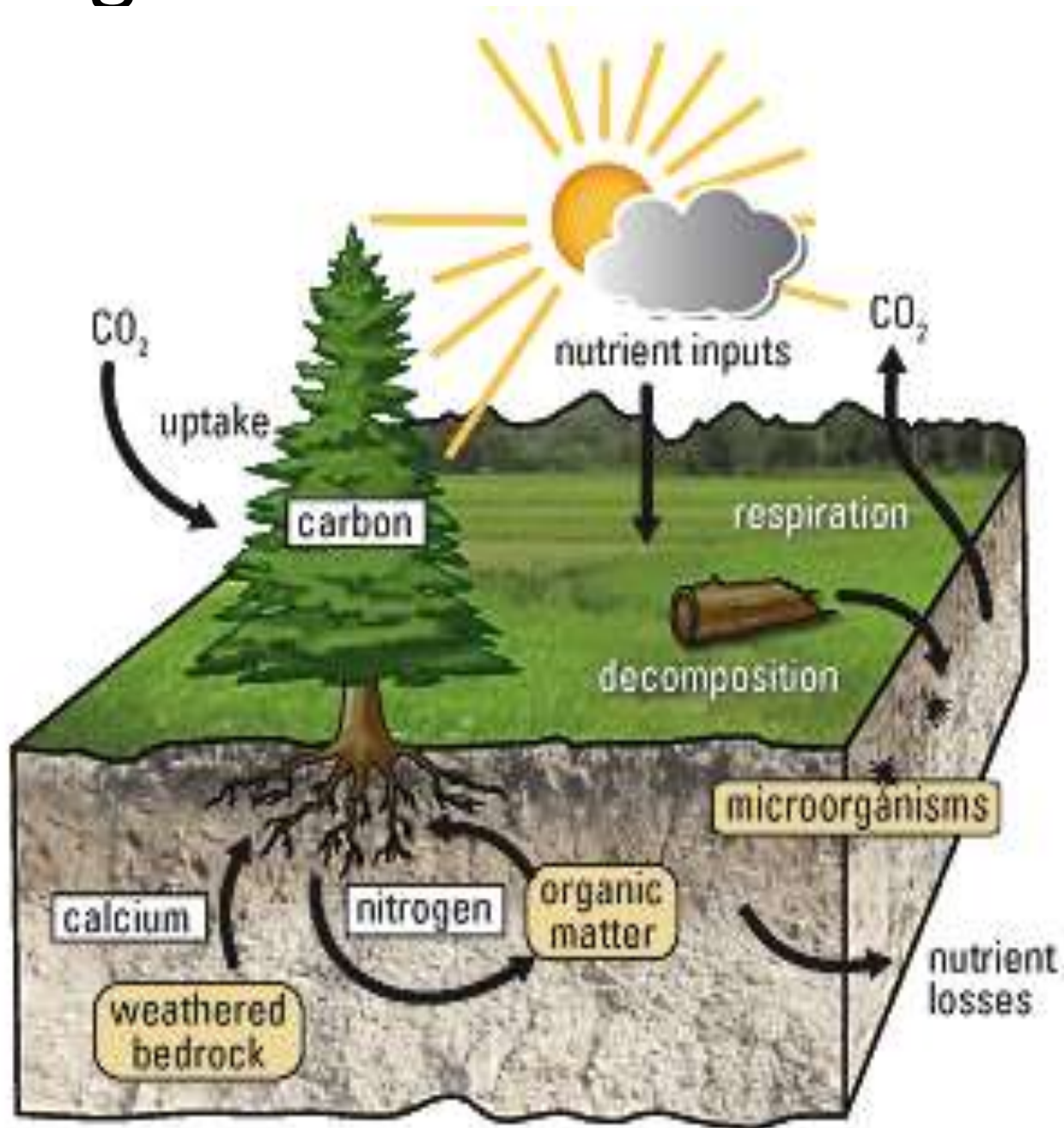
Figure 3. Nutrient concentration in plant tissues as it relates to plant performance. Below the critical range, plants are considered deficient. The range where nutrient concentration in the tissue increases, but performance does not, is considered the sufficient range. At some point beyond the sufficiency range, excessive nutrient concentrations can limit plant performance and this is considered the excessive or toxic range.



1.3.2. Soil-Plant Relations

- Beneficial aspects of soils to plants include providing physical support, heat, water, nutrients, and oxygen. Heat, light, and oxygen are also obtained by the atmosphere, but the roots of many plants also require oxygen. Nutrients, dissolved in soil water solution, are derived from soil minerals (inorganic) and organic material (see Figure 4).
- Though many aspects of soil are beneficial to plants, excessively high levels of trace metals (either naturally occurring or anthropogenically added) or applied herbicides can be toxic to some plants.
- The ratio of solids/water/air in soil is critically important to plants for proper oxygen levels and water availability. Too much porosity with air space, such as in sandy or gravelly soils, can lead to less available water to plants, especially during dry seasons when the water table is low. Too much water, in poorly drained regions, can lead to anoxic conditions in the soil, which may be toxic to almost all plants.

Figure 4. Soil-Plant Nutrient Cycle



2. NATURAL RESOURCES MANAGEMENT

2.1. Key Pillars of Resource Management

The **main parts** of natural resource management are:

- (1) Forest management/(Agro)forestry
- (2) Water resource management i.e. irrigation and water conservation/harvesting
- (3) Soil resource management and
- (4) Agriculture resource management

Forestry – which yields a vast range of products and ecosystem services useful both for local communities and at a global scale – could play a key role in accelerating a transformation towards societies that simultaneously conserve nature, better provide for human well-being, and generate income, particularly for rural people.

Forestry products can be:

- wood/timber and
- non-timber /non-wood/ forest products (NTFPs/NWFPs)]. Forests create clean air, fertile soils and rich biodiversity.

Agroforestry, an ecologically based traditional farming practice, is the use of trees and shrubs as part of agricultural systems. Planting a range of trees, perennial plants and crops to ensure vegetation stays in the ground year-round. Agroforestry generates multiple benefits, i.e. ameliorates soil fertility, controls erosion and enhances moisture retention E.g. Alley cropping is an agroforestry system in which food crops are grown in alleys between rows of hedges and it can be used as SWC practice on cultivated land.